Box-Counting Analysis of Minnesota Lake Shorelines: Lake Size, Shoreline Length, and Mapping Source

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ABSTRACT

Box-counting dimension is often used to describe how complex a shoreline is, but for mapped lakes it is an estimate over a limited range of scales, and it can depend on how each shoreline was digitized. I estimated how box counts scale between 20 and 100 m, a factor of five, for 3,274 natural lakes of at least 100 acres in the Minnesota DNR Hydrography Dataset, with two sensitivity windows. An analysis plan and decision rule were committed to version control after a pilot and before the full analysis.

The estimated exponents were close to 1, with source-group medians from 1.010 to 1.044. Most estimates below 1, 15.2% of lakes, came from a low bias of about 0.01 in how the estimator samples the boundary; rerunning with finer sampling cut that share to 4.2% and changed no conclusion. The exponent rose with shoreline length in all three windows (Spearman rho 0.41 to 0.43), meeting the planned rule, though part of this association is built into the two measures. Its correlation with lake area changed sign across windows, and two of three confidence intervals included zero; excluding the 420 narrowest lakes made it weakly positive in every window, so the size relationship depends on which lakes are included and the data show no robust relationship with lake size. Estimates differed among mapping sources (Kruskal–Wallis H = 170.90), and the sources placed vertices a median of 13 to 27 m apart; resampling every shoreline to the same vertex spacing narrowed the differences without removing them, which is consistent with digitizing contributing to the association, though geographic confounding is unresolved. Resampling spatial blocks of lakes widened confidence intervals and changed no conclusion.

1. Introduction

A lake's shoreline is where most of its exchange with the surrounding land happens. The shallow nearshore zone holds much of a lake's plant growth and fish habitat, and a more convoluted shoreline gives a lake more of that zone for its size (Wetzel, 2001). Limnologists have long summarized this with the shoreline development index, which compares a lake's shoreline length with the circumference of a circle of the same area (Hutchinson, 1957).

Shoreline length, however, depends on the scale at which it is measured. Mandelbrot (1967) showed that the measured length of a coastline grows as the measuring unit shrinks, and summarized the rate of growth with a fractal dimension D between 1 and 2: a smooth curve has D = 1, and a curve so convoluted that it fills the plane approaches D = 2. Natural features follow such scaling only over a limited range of scales, so any estimate of D belongs to the range it was

measured over (Goodchild and Mark, 1987). For lakes, this scale dependence has practical consequences. Winslow et al. (2014) estimated the total lake shoreline of the contiguous United States and showed how that total changes with the resolution of the data, mostly because coarser data miss small lakes.

This raises a question that is harder to answer than it looks: do larger lakes have more complex shorelines? The most common shape measure gives a misleading answer. Seekell, Cael and Byström (2022) showed that the shoreline development index rises with lake area for lakes of the same shape mapped at the same scale, by about 14% per doubling of area, so it cannot compare lakes of different sizes without correction. Seekell and Cael (2024) suggested that some reported ecological relationships with the index may reflect this bias. An exponent fitted over a fixed range of box sizes might avoid that size bias, but it has problems of its own: it is estimated over a limited window, small lakes span few box sizes, and statewide maps combine shorelines drawn by different agencies and methods.

Minnesota offers a useful test case. Its Department of Natural Resources (DNR) maintains a statewide map of lake polygons that records how each shoreline was drawn. In this paper I estimate box-count scaling exponents for 3,274 natural lakes and ask three questions. Does the exponent change with lake area? Does it change with shoreline length? And does it differ among mapping sources? The contribution is a regional empirical study that pays close attention to measurement: how the estimate depends on the window of box sizes, how the estimator behaves near 1, and how the estimate varies with mapping source. The analysis windows and a decision rule were fixed before the full run, and every choice made after seeing data is reported.

2. Data and methods

2.1 Data and filters

Lake polygons come from the Minnesota DNR Hydrography Dataset, layer “Dnr Hydro Features All” (Minnesota DNR, 2026), which holds 130,912 records. All lengths are measured in the layer’s own coordinate system, NAD83 / UTM zone 15N (EPSG:26915), in meters. Records were filtered in five steps (Table 1).

Table 1. Filter steps and record counts.

Step	Records kept	Before	After	Removed
1	Water-body class is “Lake or Pond”	130,912	117,235	13,677
2	Area at least 10 acres	117,235	14,640	102,595
3	Geometry present	14,640	14,640	0
4	Shoreline length at least 1,000 m	14,640	13,382	1,258
5	Entirely inside Minnesota	13,382	13,269	113

Step 1 keeps natural lakes only, so reservoirs and other water-body classes are excluded. Step 3 removed nothing, because the three records without geometry have no water-body class and were already removed in step 1. Step 5 removes lakes entirely outside Minnesota and lakes on the state border, whose polygons include shoreline outside the state.

The 13,269 lakes that pass all five steps form the exploratory sample. The primary sample is the 3,274 of them with an area of at least 100 acres. The shoreline filter removes no lake from the primary sample.

Area is the DNR acres attribute of each record. Shoreline length is the perimeter of the polygon summed over every ring, the outer shore and the shore of each island.

2.2 What is measured: a box-count scaling exponent

The boundary of each lake is every ring of its polygon, the outer shore and the shore of each island, densified so that consecutive points are no more than one-tenth of the smallest box size apart (a spacing later found to lower estimates by about 0.01; Section 3.5). For a box size s, a square grid of cells with side s is laid over the lake, and N(s) counts the cells that hold at least one boundary point. The estimate D is the least-squares slope of log N(s) against log(1/s) across the box sizes of a window:

$$\log N(s) = D \times \log(1/s) + c$$

D is a box-count scaling exponent over a finite window, not evidence that a shoreline is fractal. A stored polygon is made of straight segments, so its box-counting dimension in the limit of very small boxes is exactly 1. Over the primary window, 20 to 100 m, a factor of five, D describes how quickly the mapped boundary gains box-scale detail: D near 1 means it behaves like a smooth curve over that range, and values above 1 mean the count grows faster than 1/s. A high r^2 does not show fractal scaling, because N(s) is roughly proportional to 1/s for any smooth curve much larger than the boxes, which also gives a straight line on a log-log plot. For brevity, D is called the dimension below.

Box counts depend on where the grid starts, so the grid origin was shifted by a random fraction of s in x and y. Eight offsets were drawn with seed 0, one slope was fitted per offset, and each lake's dimension is the median slope. The same offsets were used for every lake.

Each window has 8 box sizes, evenly spaced on a log scale. The primary window runs from 20 to 100 m. Two sensitivity windows, 10 to 100 m and 20 to 200 m, were run on the same lakes. Lakes with a fit r^2 below 0.98 were counted, but none was excluded for poor fit.

The estimator was tested on shapes of known dimension: a circle (D = 1), a Koch snowflake (D ≈ 1.262) and a Sierpinski triangle (D ≈ 1.585), each passing when the estimate was within 0.05 of the known value, using boxes from 10 to 100 m. A first Koch test with a 4-iteration snowflake failed, because that shape is self-similar over too narrow a range of scales, and it was replaced with a 6-iteration snowflake. After the primary results, a review found that this snowflake was built incorrectly: its bumps pointed inward, and the polygon intersected itself, which makes it invalid geometry. The test still passed, because the box count uses only the boundary points. Section 2.5 describes the corrected benchmark and the further checks it prompted. A passing test does not by itself show that a method measures what it claims to.

2.3 Pilot, analysis plan and timeline

The design was informed by a pilot. The analysis plan was written down and committed to the project's version history (git) after the pilot and before the full analysis. It was not submitted to a registry, so it is not a formal preregistration, which the Center for Open Science defines as

specifying a research plan in advance and submitting it to a registry (Center for Open Science, n.d.). Commit times are set by the author's computer, but GitHub's own records show the plan commits were pushed at 06:28, 06:34 and 06:44 UTC on November 1, 2026, before the primary results were pushed at 06:59. The pilot box-counted 200 lakes in a 10–100 m and a 20–200 m window and found that the correlation between dimension and log area changed with the window (−0.24 and −0.43) and was strongest in small lakes. Appendix A lists the full sequence of work.

Why the primary sample starts at 100 acres. A 10-acre circle is about 230 m across, roughly twice the 100 m box and close to the 200 m box. When the largest box approaches the width of a lake, the box count levels off, which biases the estimate and can create a size trend on its own. At 100 acres, the smallest primary lake has an equivalent circular diameter of 718 m, more than seven times the largest primary box. Equivalent diameter is not width, however. A long, narrow lake, or a lake with narrow bays, can have the same area while containing sections narrower than the boxes, where both banks fall in the same box. The 100-acre cutoff reduces this problem but does not rule it out, and Section 2.5 measures actual lake widths. The switch to 100 acres was made after the pilot and before the full run, so it was informed by the data rather than fixed in advance. Its reasons were first written down after the primary results.

Why the primary window is 20 to 100 m. Under the US National Map Accuracy Standards, a 1:24,000 map places 90% of well-defined points within about 12 m (U.S. Bureau of the Budget, 1947). Shorelines are not well-defined points, so their digitizing error is likely larger, and boxes smaller than about 20 m mostly measure that error. The 100 m upper bound keeps the largest box well under the equivalent diameter of the smallest primary lake, with the caveat about width above. This window was not used in the pilot, and its reasons were also first written down after the primary results.

2.4 Statistical analysis

In the primary sample, I computed Spearman's rank correlation (Spearman, 1904) between dimension and each size measure, area and shoreline length, in each window. Each rho has a 95% percentile bootstrap interval from 2,000 resamples of lakes, with seed 0 (Efron and Tibshirani, 1993). The protocol fixed the number of resamples and the seed but not the interval method. The percentile method was recorded after the results, and BCa intervals were considered and not used. Resampling single lakes treats lakes as independent; Section 2.5 adds a spatial block bootstrap.

Decision rule (set in the analysis plan). The rule was applied separately to dimension versus area and to dimension versus shoreline length. A relationship is labeled supported only if rho has the same sign in all three windows; otherwise it is labeled window-dependent. p-values and confidence intervals are reported but are not part of the rule. The rule is kept as planned, but it is weak on its own: the three windows overlap, so their estimates are not independent, and even a negligible correlation can keep its sign in all three. Supported therefore means only that the direction was consistent across windows.

Mapping source. The DNR records how each shoreline was drawn. Sources with at least 1,000 lakes in the 10-acre sample stand alone: original USGS digital line graphs (ORIG DLG), 2004 Farm Service Agency imagery (2004 FSA), DNR 1:24,000 polygons (DNR 24k POLYS), and the Public Waters Inventory (PWI). Every other source is pooled as "other". These groups were set

on the 10-acre sample and kept unchanged for the primary sample. Dimension was compared across the five groups with a Kruskal–Wallis test (Kruskal and Wallis, 1952), in the primary window only. The 51 primary lakes with no recorded source were left out of this comparison but kept in the correlations.

Further checks. Three checks are reported alongside the decision rule but are not part of it. A within-source check repeats the correlations inside the 2004 FSA group (1,543 lakes). A sensitivity analysis drops the 100 pilot lakes that are also in the primary sample. An exploratory run uses the 10-acre sample, in the primary window only. After the results, a post-hoc check recomputed dimension on the outer shore alone, with island rings removed.

The analysis was written in Python with geopandas, shapely, numpy, scipy and pandas.

2.5 Checks added after review (not in the plan)

My reviews of the drafts, revised and expanded with ChatGPT (OpenAI), prompted the checks below, and the width results prompted the width-based exclusion. Appendix B gives their methods and full results. None was in the analysis plan, and all are labeled post-hoc.

1. **Corrected benchmarks.** The Koch snowflake is rebuilt with outward bumps, every benchmark shape is checked for valid geometry, and the benchmarks are repeated in all three windows.
2. **Calibration near 1.** The estimator is run on shapes of known dimension from 1.00 to 1.10, and on smooth shapes sized and stretched like the primary lakes, with boundary points at three spacings.
3. **Point-spacing rerun.** The full primary analysis is repeated with boundary points spaced ten times more finely.
4. **Lake width.** Actual widths are measured, replacing the equivalent-diameter argument.
5. **Spatial dependence.** Confidence intervals are recomputed by resampling spatial blocks of lakes rather than single lakes.
6. **Digitizing density.** Vertex spacing is compared across mapping sources, and the source comparison is repeated after resampling every shoreline to the same vertex spacing.
7. **Width-based exclusion.** The correlations are recomputed without the lakes narrower than 400 m at their widest point.
8. **Resampling without ring loss or invalid polygons.** The digitizing comparison is repeated on the lakes whose resampled polygons kept every ring and stayed valid.

3. Results

The estimate rose with shoreline length in every window, a consistent direction that met the planned rule, while its correlation with area changed sign across windows. Every primary lake had r^2 of at least 0.9914, but a high r^2 is expected even for smooth curves (Section 2.2), so it does not show fractal scaling. Many estimates fell below 1: 497 of 3,274 primary-window estimates (15.2%), with the lowest at 0.9667. In the sensitivity windows, 512 (15.6%) were below 1 in the 10–100 m window and 258 (7.9%) in the 20–200 m window. A polygon boundary has a limiting

dimension of exactly 1, but a slope fitted over a finite window can fall below 1 because of how boxes meet the boundary at each size and how the boundary is sampled; Section 3.5 traces most of these values to the sampling step.

3.1 Planned decision rule

Table 2. Spearman rho of dimension against shoreline length and area, primary sample (n = 3,274).

Relationship	Window	rho	95% CI	p
Shoreline length	20–100 m (primary)	0.4099	0.3806 to 0.4381	7.14e-133
Shoreline length	10–100 m	0.4261	0.3975 to 0.4537	1.43e-144
Shoreline length	20–200 m	0.4240	0.3968 to 0.4514	4.73e-143
Area	20–100 m (primary)	−0.0048	−0.0388 to 0.0291	0.784
Area	10–100 m	0.0197	−0.0141 to 0.0535	0.259
Area	20–200 m	−0.0590	−0.0942 to −0.0247	0.000725

- **Dimension versus shoreline length: supported.** rho is positive in all three windows (Figure 1). Under the planned rule, this means only that the direction was consistent.
- **Dimension versus area: window-dependent.** rho is negative in the 20–100 m and 20–200 m windows and positive in the 10–100 m window (Figure 2). Two of the three intervals include zero; only the 20–200 m interval excludes it.

3.2 Mapping source

Dimension differed across the five source groups (Kruskal–Wallis H = 170.90, p = 6.71e-36, 3,223 lakes). The group medians span 0.034, from 1.0100 to 1.0444 (Table 3, Figure 3). This is an association between source and estimate. It does not show that the mapping method caused the difference (Section 4). The original validation tolerance of 0.05 was wider than this span; the post-review calibration measured the estimator's error more finely, and the differences persisted when the analysis was rerun with finer boundary sampling (Section 3.5).

Table 3. Median primary-window dimension by mapping source.

Source group	Lakes	Median dimension
PWI	54	1.0444
2004 FSA	1,543	1.0289
Other	1,110	1.0242
ORIG DLG	293	1.0122
DNR 24k POLYS	223	1.0100

3.3 Checks reported alongside the rule

Within 2004 FSA (1,543 lakes). rho with shoreline length was positive in all three windows (0.4030 to 0.4368). rho with area was negative in all three windows, from −0.0689 to −0.1018, and none of the three intervals included zero (primary window: −0.0834, 95% CI −0.1338 to −0.0320).

Pilot lakes excluded (3,174 lakes). Every rho kept the sign it had in the full primary sample. Shoreline rho ranged from 0.4182 to 0.4360. For area, as in the full sample, the 20–100 m and 10–100 m intervals included zero and the 20–200 m interval did not (rho −0.0565, 95% CI −0.0898 to −0.0201).

3.4 Exploratory and post-hoc results

These results were not in the analysis plan, and no decision rule was applied to them.

Exploratory 10-acre sample (13,269 lakes, primary window). rho with area was −0.2831 (95% CI −0.2988 to −0.2676), and rho with shoreline length was 0.0711 (95% CI 0.0549 to 0.0874). One lake had r^2 below 0.98. The area correlation is far stronger than in the primary sample (−0.2831 against −0.0048), in the size range where finite-size bias and the shoreline filter are expected to matter (Sections 2.3 and 5).

Post-hoc outer-shore check. Of the 3,274 primary lakes, 1,135 have at least one island. With island rings removed, rho between outer-shore dimension and outer-shore length was 0.3646 (95% CI 0.3342 to 0.3942), down from 0.4099. rho with area was −0.0244 (95% CI −0.0589 to 0.0090).

3.5 Post-review checks

Eight checks were added after review (Section 2.5). Table 4 summarizes them, and Appendix B gives their methods and full results. The planned conclusions held under every check except the two width-based exclusions: both made the area correlation positive in every window, and both select partly on the estimate, the narrow-shore subset almost entirely. The most consequential finding was a bias in the planned estimator. Points spaced at a tenth of the smallest box made smooth shapes read about 0.01 low, which accounts for most estimates below 1; rerunning the primary analysis with ten times finer spacing cut the share below 1 from 15.2% to 4.2% and moved no correlation by more than 0.0035.

Table 4. Checks added after review and what each found.

Check	Main result	Effect on the planned conclusions
Corrected benchmarks	The valid 7-iteration Koch snowflake came within 0.016 of its known dimension in every window	None
Calibration near 1	At the planned point spacing, smooth shapes read about 0.01 low; with finer spacing, known dimensions of 1.02 and 1.05 were recovered within 0.001	None
Point-spacing rerun	Estimates below 1 fell from 15.2% to 4.2% of lakes; no correlation changed by more than 0.0035	None
Lake width	Only 3 lakes are narrower than 200 m at their widest; the narrow-shore share tracked the estimate (rho 0.86 to 0.87)	Its subset made area positive but is not interpretable
Spatial block bootstrap	Intervals widened at 25, 50 and 100 km blocks; no primary interval changed whether it includes zero	None
Digitizing density	Median vertex spacing ranged from 13 to 27 m by source; resampling to equal spacing narrowed the source gaps without removing them	None
Width-based exclusion	Without the 420 narrowest lakes, the area correlation was weakly positive in every window (0.08 to 0.13)	Area becomes supported in a subset that partly selects on the estimate
Resampling without ring loss or invalid polygons	On the 2,662 lakes that kept every ring and a valid polygon, resampling still roughly halved the gap between 2004 FSA and ORIG DLG, from 0.0087 to 0.0044; every interval excluded zero	None

3.6 Figures

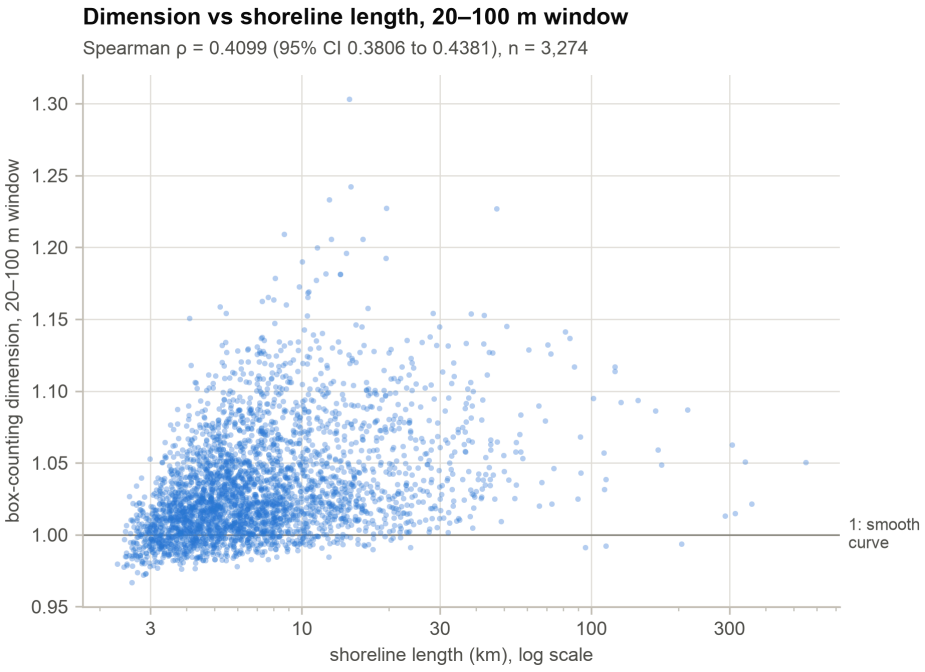


Figure 1. Primary-window dimension against shoreline length (km, log scale), one point per primary lake. A gray line marks D = 1, the dimension of a smooth curve.

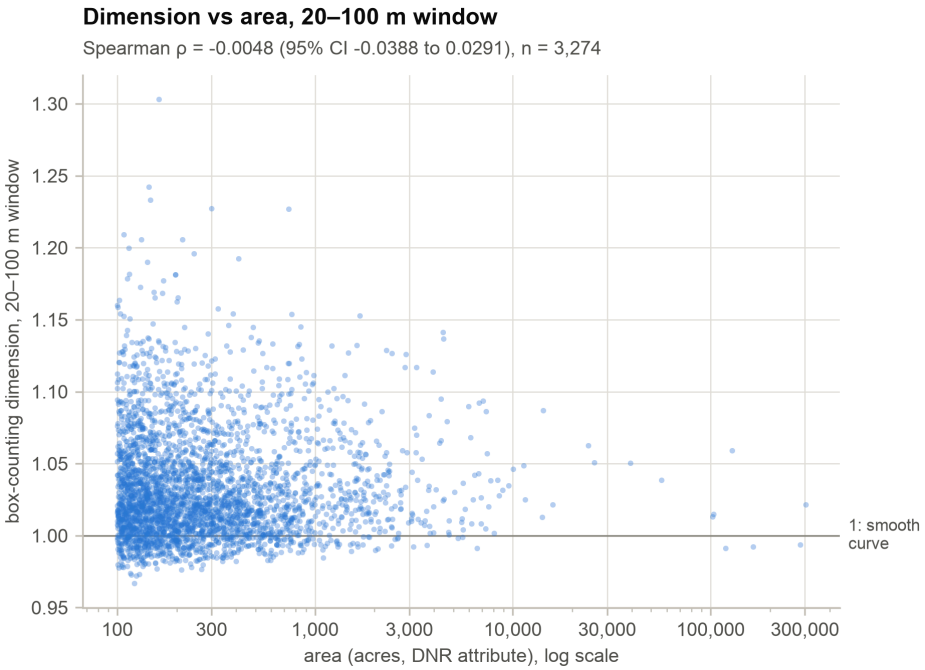


Figure 2. Primary-window dimension against area (acres, log scale), one point per primary lake.

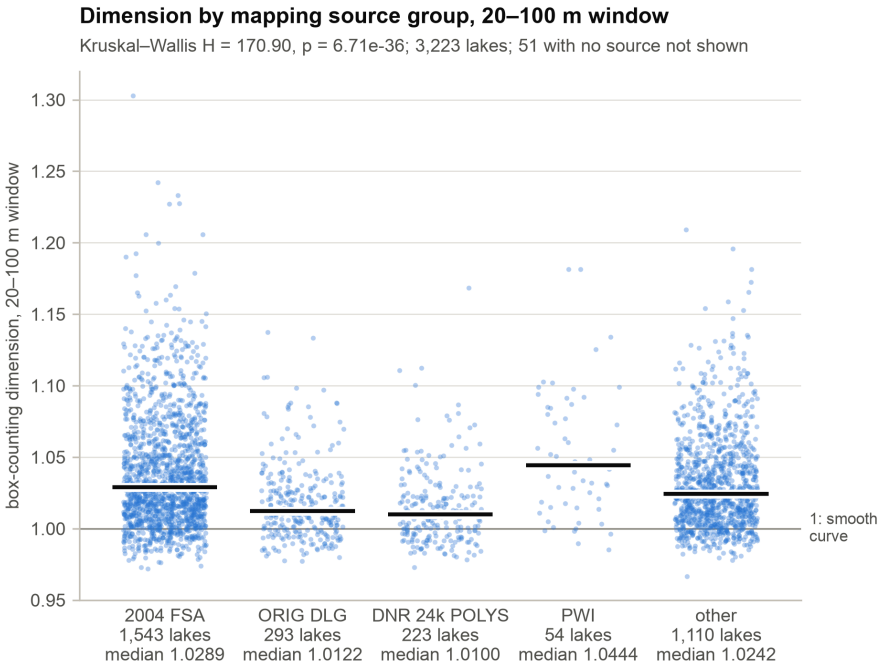


Figure 3. Median primary-window dimension by mapping source group, drawn over every lake in the group. Lakes with no recorded source are not shown.

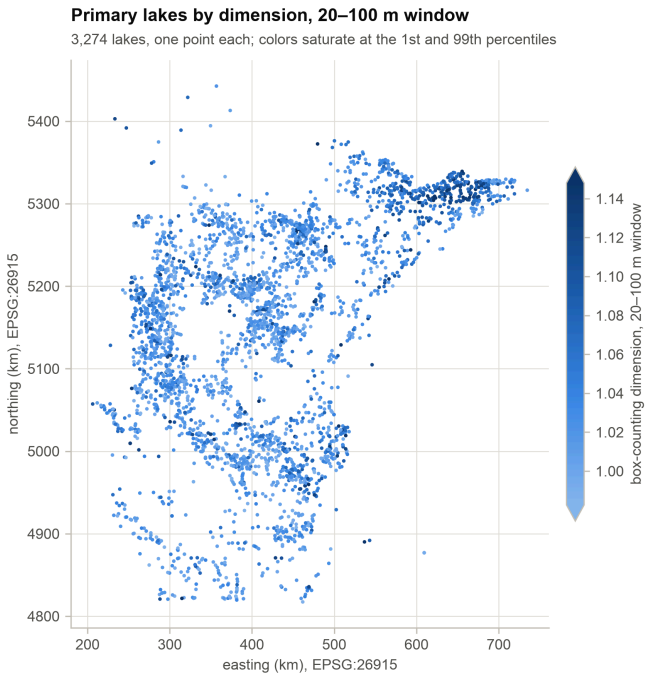


Figure 4. Map of the primary lakes, each drawn as one point and colored by primary-window dimension. Colors saturate below the 1st and above the 99th percentile.

4. Discussion

Among Minnesota lakes of at least 100 acres, the box-count exponent showed no consistent relationship with lake size, rose with shoreline length, and differed among mapping sources. Each finding needs a careful reading, starting with what was measured.

Most estimates below 1 came from the estimator. Median estimates ran from 1.010 to 1.044, and 15.2% were below 1. That is not a contradiction: a slope fitted over a factor-of-five window is not the limiting dimension, and finite-window and numerical effects can push it below 1. The calibration traced most of this to how the estimator samples the boundary: points spaced at a tenth of the smallest box miss some boxes the shoreline only clips, which lowered smooth-shape estimates by about 0.01. With finer spacing, the share below 1 fell to 4.2% and no conclusion changed. The remaining error depends on shape: narrow small shapes read high in the widest window, and a regular test fractal was off by up to 0.015 over a factor-of-five window. Differences of a few hundredths between groups of lakes are therefore close to what this estimator can resolve. What the data do show is that mapped shorelines gain little box-scale detail between 20 and 100 m; whether natural shorelines are fractal at finer scales cannot be tested with statewide maps.

Lake size. The relationship with area depends on which lakes are included. It was clearly negative in the 10-acre exploratory sample (−0.28), close to zero and mixed in sign in the primary sample, and weakly positive in all three windows (0.08 to 0.13) once the 420 lakes narrower than 400 m at their widest were excluded. Those narrow lakes have higher estimates: across all primary lakes the estimate falls as width rises, most steeply in the 20–200 m window, where the calibration found narrow shapes read high. Some of their excess is probably that box-size effect, but a lake is often narrow because bays and arms cut into it, which raises the estimate for

real, and these data cannot separate the two. Within 2004 FSA, the area correlation was small but negative in all three windows. The defensible reading is that the size relationship is fragile, not that larger or smaller lakes have more complex shorelines. That still differs from the shoreline development index, which rises with area even for lakes of the same shape (Seekell, Cael and Byström, 2022). In the calibration, smooth shapes of the same proportions scored within about 0.006 of each other from 100 to 10,000 acres, except the narrowest small one, so the box-count exponent has no comparable built-in size trend.

Shoreline length. Dimension rose with shoreline length in every window and within the 2004 FSA group, and removing islands lowered the correlation only from 0.41 to 0.36. Part of this result is built into the measures. For lakes of the same area, a longer shoreline is a more convoluted one, and box counts at 20 to 100 m respond to the same convolution. The finding is best read as a consistency check: the estimate tracks how convoluted a shoreline is, not how large the lake is.

Mapping source is associated with the estimate. The spread of median estimates across source groups, 0.034, is as large as the effects this study looked for. It persisted under the finer-sampling rerun and under spatial block resampling, so it is not the point-spacing bias or an artifact of treating clustered lakes as independent. Block resampling does not separate mapping source from regional geology, however, so geographic confounding remains unresolved. The evidence is consistent with digitizing contributing to the difference: the sources traced shorelines with vertices a median of 13 to 27 m apart, denser vertices went with higher estimates within every source, and resampling every shoreline to the same spacing narrowed the gaps between sources, nearly halving the gap between 2004 FSA and ORIG DLG, without removing them. Resampling also dropped small rings and left a few polygons invalid, but on the 2,662 lakes that kept every ring and a valid polygon, it still roughly halved that gap, from 0.0087 to 0.0044 (Appendix B). The clean subset also shows a second part of the gap. The lakes left out, nearly all because a ring of 150 m or less was dropped, make up 22.9% of 2004 FSA but 2.0% of ORIG DLG, and without them the gap between the two is about half as large before any resampling, 0.0087 against 0.0167. Neither result shows that the differences are errors: resampling removes real shoreline detail finer than its interval along with tracing detail, and 2004 FSA may have recorded small islands that the maps left out, or simply covered lakes that have more of them. The rest of the difference could reflect real differences among the lakes each source covers, in location, geology or lake type, or digitizing choices that vertex spacing does not capture, such as how closely each tracer followed the water's edge. The direct test is still to compare the same lakes mapped by different methods, for example by tracing a sample of lakes from current aerial imagery. Until then, comparisons of shoreline complexity across many lakes should record how each shoreline was digitized, including its vertex spacing, and check whether results hold within a single source.

A spatial pattern. The map of dimensions (Figure 4) shows higher values in the northeast of the state. Two explanations fit and were not tested here. Many northeastern lakes sit in glacially scoured bedrock basins, which could give them more irregular shores. But mapping sources also vary by region, and source is associated with the estimate on its own.

Next steps. Four follow-ups would address the main limits of this study. The first is a same-lake comparison: the same lakes mapped by two methods, which would separate the mapping effect from geography. The second is high-resolution shorelines, for example from lidar elevation data, to reach scales below 20 m. The third is to relate the estimate to shoreline length while holding area fixed, through a bias-corrected shoreline development index or a partial correlation. The fourth is procedural: a follow-up study should submit its plan to a public registry before any data are analyzed.

Conclusion. Box-count scaling exponents estimated from Minnesota's statewide lake map are close to 1, depend on the window of box sizes, and differ among mapping sources, a pattern consistent with a contribution from how densely the sources digitized shorelines, though source cannot yet be separated from geography. Among lakes of at least 100 acres, they show no robust link to lake size. Studies that compare shoreline complexity across many lakes should fix their scale range in advance, calibrate the estimator near 1, and record how densely each shoreline was digitized.

5. Limitations

1. **Shoreline length partly encodes complexity.** Shoreline length and the estimate both respond to how convoluted a shoreline is, so part of the positive relationship is built in. No analysis held area fixed, and none was planned.
2. **Narrow parts cannot be separated from shoreline detail.** Measured widths show that whole lakes rarely approach the box size in the primary window: only 3 lakes are narrower than 200 m at their widest. But 420 are narrower than 400 m, twice the largest box of the 20–200 m window, where a 100-acre ellipse 227 m wide read 0.035 high in the calibration, and the only primary area correlation whose interval excludes zero (−0.059) comes from that window. Most lakes also have bays and inlets narrower than the largest box, and the measure of those narrow parts tracked the estimate almost perfectly, so narrowness at the box scale cannot be separated from the shoreline detail the estimate measures. Excluding the 420 narrowest lakes turned the area correlation positive in every window, but width also correlates with the estimate (rho −0.24 to −0.37), so that exclusion selects partly on it, and the size result depends on which lakes are included.
3. **The estimator's error is close to the size of the effects.** As planned, the estimator read about 0.01 low on smooth shapes because of how it samples the boundary, which explains most of the 15.2% of estimates below 1. Finer sampling removed most of that bias and changed no conclusion, but the remaining error still depends on shape: up to 0.015 for a regular test fractal in the primary window, and 0.035 to 0.043 for a narrow small shape in the 20–200 m window. The calibration used synthetic shapes, not real shorelines from each source, so it does not give the error for any particular lake. The Koch benchmark was invalid geometry until after the results.
4. **The shoreline filter selects on shape.** A circular 10-acre lake has less than 1,000 m of shoreline, so step 4 keeps a small lake only if its shoreline is long for its area. This favors convoluted small lakes and may push the exploratory area correlation toward negative values. It removes no lake from the primary sample.

5. **The 20 m lower bound sits close to digitizing error.** The 20 m floor is less than twice the 12 m map accuracy standard, and shoreline error is likely larger, so the smallest primary boxes may still pick up digitizing error. Every window starts at 20 m or below, so no window tests this. The bound also assumes map-based digitizing, but 1,543 primary lakes, the 2004 FSA group, were traced from aerial imagery, so its reasoning fits them less exactly.
6. **Lakes are not independent.** Lakes cluster in space, share regional geology and were mapped in regional projects. Resampling spatial blocks widened the intervals and changed no primary conclusion, but it only approximates this dependence: the 100 km blocks were too few and uneven to be reliable, and no spatial model was fitted. The p-values still assume independence.
7. **No correction for multiple comparisons.** The primary sample alone gives six correlations, and the checks add more. Every p-value is unadjusted. The decision rule uses only the sign of rho, so it is unaffected.
8. **The decision rule is weak.** A consistent sign across three overlapping windows shows directional consistency only, and even a negligible correlation can meet it.
9. **The analysis plan was not registered.** The plan was committed to the project's version history after the pilot, not submitted to a registry. GitHub's push records confirm it was pushed before the results, but they are not a registry entry. The 100-acre primary sample and the 20–100 m window were chosen after the pilot, and their reasons were written down only after the results.
10. **Mapping source is confounded with location.** Resampling to equal vertex spacing narrowed the source differences, including on the lakes that kept every ring and a valid polygon, but resampling removes real shoreline detail finer than its interval along with tracing detail, so it does not isolate vertex spacing. Those lakes are also a smaller share of 2004 FSA (77.1%) than of ORIG DLG (98.0%) or DNR 24k POLYS (99.1%). The differences that remain cannot be split between other digitizing differences and real, regional differences among the lakes each source covers. Spatial block resampling addresses uncertainty from clustering, not this confounding.
11. **Post-review checks add forking paths.** Eight checks were added after the reviews, and three of them leave out lakes by shape, which selects partly on the estimate: the narrow-shore subset almost entirely, so it is not interpreted; the width-based exclusion more weakly; and the clean subset of the resampling check, which leaves out lakes that lost a ring or became invalid in resampling, most of them in 2004 FSA, so its main use is to compare the same lakes before and after resampling. Each check is reported and labeled post-hoc.

Data and code availability

All code, tests, the analysis plan and its full revision history are in the public repository github.com/Ntalatam/lake-fractals, archived on Zenodo at doi.org/10.5281/zenodo.23997837 (all versions); version 1.0.0 (doi.org/10.5281/zenodo.23997838; Talatam, 2026) holds every result reported here. Every result in this paper is written by a script to a results file in that repository. On the original computer, a fresh copy of the repository reproduced every output of the fast steps, identical apart from recorded run times; the three longest runs were not repeated, and the clean-subset check was added after that run. The tests compare box counts exactly and

other values to within 1e-12, because floating-point results can differ in their last digits on other systems; byte-for-byte checks against the original outputs run only on request. The lake data are provided by the Minnesota Department of Natural Resources (MNDNR) and used under its General Data and Software License Agreement, which requires acknowledging MNDNR and does not allow the complete dataset to be redistributed. The repository's README gives the exact source, the download steps and a checksum of the file used. Code is released under the MIT License, and the paper, its text and figures under CC BY 4.0.

Use of AI tools

I directed this study and made the final decisions about its scope, analytical methods, and interpretation. I used Claude (Anthropic) and ChatGPT (OpenAI) to assist with generating and revising code, developing tests, evaluating methodological choices, and drafting and revising the manuscript. I reviewed the analyses and proposed revisions, determined which approaches and conclusions to retain, and take responsibility for the reported results, their interpretation, and the limitations described in this paper.

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Appendix A. Analysis timeline

The sequence of work, from the project's git history:

1. **Initial plan.** The main sample was lakes of at least 10 acres, with 100 acres as a robustness check.
2. **Pilot.** 200 lakes, 50 drawn at random from each of four acreage bins, were box-counted in a 10–100 m and a 20–200 m window. The correlation between dimension and log area was −0.24 and −0.43 in the two windows. The relationship therefore changed with the window and was strongest in small lakes.
3. **Analysis plan.** Before any full-sample dimension was computed, I recorded the 100-acre primary sample, the 20–100 m primary window, the two sensitivity windows, the decision rule, the bootstrap intervals, a within-source check, a pilot-excluded sensitivity analysis, and the mapping-source groups.
4. **Primary results.** The full run followed.
5. **After the results.** I added a note on the bootstrap method, a post-hoc outer-shore check, disclosures found in an audit of the code and results by a separate Claude Code session, and the written reasons for the sample and window choices in Section 2.3.
6. **Post-review checks.** Reviews of the drafts led to the eight checks in Section 2.5, added on November 1 and 2, 2026. The primary results files have not changed since they were first committed.

Appendix B. Checks added after review

B.1 Methods

Benchmarks. The Koch snowflake now has outward bumps and 7 iterations at a side of 7,290 m, so its smallest segment, 3.3 m, is at most a third of the smallest box in every window, and the whole shape is more than ten times the largest box. Each benchmark shape is checked for valid geometry, and the snowflake's area is checked against its known value, 8/5 of the starting triangle, which the inward build fails. The circle (radius 2,000 m), the 6- and 7-iteration snowflakes and the Sierpinski triangle (8 levels, side 1,280 m) were run in all three windows. The 0.05 tolerance is kept only as a guard against code changes, not as a measure of accuracy.

Calibration. Generalized Koch snowflakes with known D of 1.02, 1.05 and 1.10 were built by narrowing the angle of each bump, with 6 iterations at a side of 7,290 m. The D = 1.10 snowflake's smallest segment, 3.8 m, is slightly more than a third of the 10 m box, so it misses the sizing rule in the 10–100 m window. Smooth ellipses of 100, 1,000 and 10,000 acres, with length-to-width ratios of 1, 3 and 10 and vertices every 10 m, were each run at 40 random rotations; their true D is 1. Random shifts were also applied but cannot change any count, because the grid is placed relative to each shape's lower-left corner. Every shape was run in all three windows with boundary points spaced at 0.1 of the smallest box, as planned, and at 0.03 and 0.01.

Point-spacing rerun. The full primary analysis was repeated on the same 3,274 lakes with points spaced at 0.01 of the smallest box, with every planned statistic and the decision rule applied as written. The planned results remain the main results, because switching estimators af-

ter seeing results would itself be a change to the plan.

Lake width. For each primary lake, the widest point was measured as the diameter of the largest circle that fits inside the polygon, to within 1 m, along with the short side of the smallest rotated rectangle around it. To measure narrow parts, each lake was opened: shrunk inward by a distance r and grown back by r, which removes any part narrower than 2r. The narrow-shore share is the fraction of shoreline lying more than 1 m from the opened lake, with r = 50 m for the 100 m box and r = 100 m for the 200 m box. One sensitivity check, set before its results were seen, dropped lakes whose share for a window's largest box exceeded 0.10 and recomputed the correlations.

Spatial dependence. Each primary lake was assigned by its centroid to a square block of 25, 50 or 100 km. Blocks were resampled with replacement, 2,000 times with seed 0, to give percentile intervals for the correlations and the source-group medians. For the 2004 FSA correlations and each group median, blocks were resampled among that group's own lakes.

Digitizing density. Vertex spacing, the distance between consecutive vertices as stored, was summarized for each lake by its median. To check whether vertex density could account for part of the source differences, every ring was rebuilt with vertices every 25 m and every 50 m along its length. Rings too short to keep 4 distinct vertices, those of 75 m or less at 25 m and 150 m or less at 50 m, were dropped, along with any separate part of a lake whose outer ring was dropped. The primary-window estimate was recomputed for every lake and the source comparison repeated. Because resampling also dropped rings and left some polygons invalid, the comparison was then repeated on the clean subset, the lakes whose resampled polygons kept every ring and stayed valid at both spacings. On it and on all lakes, the gaps between 2004 FSA and the two map-based groups were reported directly, each with a 95% percentile interval from 2,000 resamples of lakes within each group (seed 0), alongside H. No lake was box-counted again for this check.

Width-based exclusion. Because the narrow-shore share turned out to track the estimate closely (Section 3.5), a final check, set before its results were seen, uses each lake's widest point instead, a plain geometric measure that does not count small shoreline detail. The six primary correlations and the decision rule are recomputed without the lakes narrower than 400 m at their widest point, twice the largest box of the 20–200 m window, dropping the same lakes in all three windows, and the estimate is correlated with that width in each window. The computed width is accurate to within 2 m.

Every check that uses lake estimates was run with both the planned estimates and the finer-spacing estimates.

B.2 Results

Benchmarks. Every benchmark shape is now valid geometry. At the planned point spacing, the corrected 7-iteration snowflake came within 0.016 of its known value in every window (Table B1). The 6-iteration snowflake, now valid, read 0.034 low in the 10–100 m window, close to the 0.05 tolerance the invalid version had passed.

Table B1. Benchmark estimates at the planned point spacing (bias from the known value in parentheses).

Shape (known D)	10–100 m	20–100 m	20–200 m
Circle (1.000)	0.9913 (−0.009)	0.9863 (−0.014)	0.9934 (−0.007)
Koch, 6 iterations (1.262)	1.2281 (−0.034)	1.2489 (−0.013)	1.2524 (−0.010)
Koch, 7 iterations (1.262)	1.2468 (−0.015)	1.2556 (−0.006)	1.2587 (−0.003)
Sierpinski (1.585)	1.5706 (−0.014)	1.5765 (−0.008)	1.5752 (−0.010)

Calibration. At the planned spacing, the estimator read low on smooth shapes in the primary window: 0.009 to 0.014 low for the circle and for 8 of the 9 ellipse types, and 0.001 low for the narrowest small ellipse (Table B2). Spacing points ten times more finely removed most of this bias: smooth shapes then came within 0.002 of 1, except the narrowest small ellipse, at 1.009. Known dimensions of 1.02 and 1.05 were recovered within 0.001 at the finer spacing. The D = 1.10 snowflake read 0.010 high at the planned spacing and 0.015 high at the finer one. Because that error grew as the spacing became finer, it comes from the test shape's regular, stepwise structure over a short window, not from point spacing.

Table B2. Calibration in the primary window (20–100 m): mean estimate at two point spacings.

Shape	Known D	Planned spacing (0.1)	Finer spacing (0.01)
Circle, radius 2,000 m	1.00	0.9863	0.9984
Smooth ellipses, 9 sizes and shapes, 40 rotations each	1.00	0.9857 to 0.9986	0.9985 to 1.0089
Generalized Koch	1.02	1.0155	1.0200
Generalized Koch	1.05	1.0461	1.0494
Generalized Koch	1.10	1.1099	1.1153

Narrow shapes. The narrowest small shape, a 100-acre ellipse 227 m wide, read 0.035 high in the 20–200 m window at the planned spacing and 0.043 high at the finer spacing. When the largest boxes approach a lake's width, both shores fall in the same boxes, and the estimate rises.

Estimates below 1. Most estimates below 1 came from the point-spacing bias. With points spaced at 0.01 of the smallest box, the share below 1 in the primary window fell from 15.2% to 4.2% (Table B3).

Table B3. Lakes with an estimate below 1, out of 3,274.

Window	Planned spacing (0.1)	Finer spacing (0.01)
20–100 m (primary)	497 (15.2%), lowest 0.9667	138 (4.2%), lowest 0.9742
10–100 m	512 (15.6%), lowest 0.9768	110 (3.4%), lowest 0.9843
20–200 m	258 (7.9%), lowest 0.9779	74 (2.3%), lowest 0.9869

Point-spacing rerun. No conclusion changed. Area remained window-dependent (rho −0.0066, 0.0203 and −0.0585 in the 20–100, 10–100 and 20–200 m windows), and shoreline length remained supported (0.4106, 0.4267 and 0.4250). Across the primary, within-2004 FSA and pilot-excluded samples, all 18 correlations kept their sign, every interval kept its status of including

or excluding zero, and the largest change in rho was 0.0035. Every source-group median rose by 0.0085 to 0.0099, the groups kept their order, and the Kruskal–Wallis test gave H = 164.98 (p = 1.25e-34), against 170.90 as planned. The bias shifted nearly all lakes by a similar amount, which is why the rank correlations barely moved.

Lake width. At their widest point, primary lakes ranged from 166 to 23,010 m across (median 610 m; 5th percentile 341 m). Only 3 lakes were narrower than 200 m, twice the largest box of the primary window, and 420 (12.8%) were narrower than 400 m, twice the largest box of the 20–200 m window. Narrow parts were common, however: the median share of shoreline in parts narrower than the largest box was 0.185 for the 100 m box and 0.434 for the 200 m box.

The narrow-shore share turned out to measure almost the same thing as the estimate (rho 0.86 to 0.87 in every window), because opening a lake also removes bays, points and other shoreline detail smaller than the box. Dropping lakes with a share above 0.10 kept only 745 lakes in the 10–100 and 20–100 m windows and 98 in the 20–200 m window. In those subsets the area correlation was positive in all three windows (0.18 to 0.21), which the decision rule would label supported. Because the subsets were chosen on a measure this close to the estimate, they keep mostly the smoothest shorelines, and a correlation computed after that kind of selection is not reliable evidence about lake size. The result is reported but not interpreted.

Width-based exclusion. Dropping the 420 lakes narrower than 400 m at their widest point left 2,854 lakes, the same in all three windows (Table B4). Shoreline-length correlations barely changed, at 0.41 to 0.44. The area correlation became positive in all three windows, at 0.099, 0.127 and 0.076 in the 20–100, 10–100 and 20–200 m windows, with every interval excluding zero, so the decision rule would label it supported; the finer-spacing estimates gave the same result. Width is not independent of the estimate, however. Across all 3,274 lakes, the estimate fell as width rose (rho −0.27, −0.24 and −0.37 in the 20–100, 10–100 and 20–200 m windows), most steeply in the window where the calibration found narrow shapes read high. This exclusion therefore also selects partly on the estimate, though far less than the narrow-shore share. Eighteen excluded lakes lie within 2 m of the cutoff, the error margin of the width calculation.

Table B4. Spearman rho on the 2,854 lakes at least 400 m wide at their widest point (planned estimates).

Relationship	Window	rho	95% CI
Shoreline length	20–100 m	0.4108	0.379 to 0.440
Shoreline length	10–100 m	0.4295	0.399 to 0.459
Shoreline length	20–200 m	0.4368	0.407 to 0.466
Area	20–100 m	0.0991	0.063 to 0.135
Area	10–100 m	0.1266	0.091 to 0.164
Area	20–200 m	0.0757	0.040 to 0.112

Spatial dependence. The primary lakes fell into 273, 91 and 30 blocks at 25, 50 and 100 km. Resampling blocks instead of lakes widened the intervals but changed no primary-sample interval's status of including or excluding zero, at any block size (Table B5). Within 2004 FSA, one interval changed: the 10–100 m area correlation included zero at 50 km. The 100 km results are the least reliable: there are only 28 to 30 blocks, holding from 1 to 326 lakes each, and

within 2004 FSA all three 100 km area intervals came out narrower than the lake-level ones. Source-group medians also had wider intervals, but 2004 FSA and the pooled other group stayed separate from ORIG DLG and DNR 24k POLYS at every block size.

Table B5. Spearman rho in the primary sample with 95% intervals from resampling lakes or spatial blocks (planned estimates).

Relationship	Window	rho	Lakes	25 km blocks	50 km blocks	100 km blocks
Shoreline length	20–100 m	0.4099	0.381 to 0.438	0.368 to 0.450	0.358 to 0.467	0.353 to 0.475
Shoreline length	10–100 m	0.4261	0.398 to 0.454	0.386 to 0.464	0.375 to 0.478	0.374 to 0.488
Shoreline length	20–200 m	0.4240	0.397 to 0.451	0.384 to 0.464	0.373 to 0.479	0.363 to 0.492
Area	20–100 m	−0.0048	−0.039 to 0.029	−0.043 to 0.034	−0.044 to 0.039	−0.039 to 0.034
Area	10–100 m	0.0197	−0.014 to 0.054	−0.019 to 0.059	−0.019 to 0.062	−0.014 to 0.055
Area	20–200 m	−0.0590	−0.094 to −0.025	−0.096 to −0.022	−0.099 to −0.013	−0.095 to −0.020

Digitizing density. The mapping sources placed vertices at very different densities, from a median of 13.2 m apart in 2004 FSA to 27.4 m in ORIG DLG (Table B6; Kruskal–Wallis H = 642.56). Denser vertices went with higher estimates, both overall (rho −0.52 between vertex spacing and the estimate) and within every source (−0.30 to −0.60). Correlation alone cannot tell which way this runs, since a more convoluted shoreline may also be traced with more vertices. Resampling every shoreline to vertices every 25 m or 50 m lowered every group's median and narrowed the differences between groups without removing them. At 50 m, the Kruskal–Wallis statistic fell from 170.90 to 114.27 (from 164.98 to 108.25 at the finer point spacing), and the gap between 2004 FSA and ORIG DLG shrank from 0.0167 to 0.0091; every p-value stayed below 1e-21. Resampling dropped 1,959 of 10,836 rings at 25 m, from 350 lakes, and 3,373 at 50 m, from 603 lakes, and it left 9 and 24 lakes with invalid polygons, which were kept because box counting uses only boundary points. Because resampling also dropped rings and left some polygons invalid, it does not on its own isolate vertex density, and a fall in H is not a share of the differences explained.

Table B6. Vertex spacing and median primary-window estimate by mapping source, as digitized and after resampling (planned point spacing).

Source group	Lakes	Median vertex spacing (m)	As digitized	Vertices every 25 m	Vertices every 50 m
2004 FSA	1,543	13.2	1.0289	1.0216	1.0159
Other	1,110	14.2	1.0242	1.0181	1.0127
PWI	54	16.9	1.0444	1.0372	1.0279
DNR 24k POLYS	223	19.2	1.0100	1.0053	1.0032
ORIG DLG	293	27.4	1.0122	1.0092	1.0068
Kruskal–Wallis H across groups			170.90	138.52	114.27

Resampling without ring loss or invalid polygons. Resampling kept every ring and a valid polygon at both spacings for 2,662 of the 3,274 lakes, the clean subset; of the other 612, 603 lost a ring at 50 m and the rest were left invalid. On the clean subset, resampling still narrowed the differences between sources (Table B7). The gap between 2004 FSA and ORIG DLG fell from 0.0087 as digitized to 0.0063 at 25 m and 0.0044 at 50 m, and the gap between 2004 FSA and DNR

24k POLYS fell from 0.0108 to 0.0080; at the finer point spacing, the first gap fell from 0.0095 to 0.0038. Every gap's interval excluded zero, in both subsets and at both point spacings, the groups kept the same order of medians in all twelve combinations of subset, point spacing and vertices, and the largest Kruskal–Wallis p-value was 1.68e-10. The clean subset is not a random sample of the sources, however. It keeps 77.1% of 2004 FSA lakes but 98.0% of ORIG DLG and 99.1% of DNR 24k POLYS lakes, and leaving out the rest lowers the 2004 FSA median from 1.0289 to 1.0207 while the map-based medians barely move. On the clean subset, the gap between 2004 FSA and ORIG DLG is therefore already about half the all-lakes gap before any resampling, 0.0087 against 0.0167.

Table B7. Gaps between source-group medians and Kruskal–Wallis H on the clean subset and on all lakes (planned point spacing; 95% intervals in parentheses).

Comparison	Vertices	Clean subset (2,625 lakes)	All lakes (3,223 lakes)
2004 FSA − ORIG DLG	As digitized	0.0087 (0.0050 to 0.0118)	0.0167 (0.0127 to 0.0201)
	Every 25 m	0.0063 (0.0021 to 0.0100)	0.0124 (0.0082 to 0.0163)
	Every 50 m	0.0044 (0.0018 to 0.0067)	0.0091 (0.0062 to 0.0117)
2004 FSA − DNR 24k POLYS	As digitized	0.0108 (0.0078 to 0.0160)	0.0189 (0.0152 to 0.0241)
	Every 25 m	0.0102 (0.0074 to 0.0138)	0.0164 (0.0133 to 0.0203)
	Every 50 m	0.0080 (0.0058 to 0.0122)	0.0127 (0.0106 to 0.0170)
Kruskal–Wallis H	As digitized	88.72	170.90
	Every 25 m	69.50	138.52
	Every 50 m	56.49	114.27

No planned conclusion changed under the spatial or digitizing checks.